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Managing Subsurface Uncertainty Across the Asset Lifecycle with OSDU RDDMS

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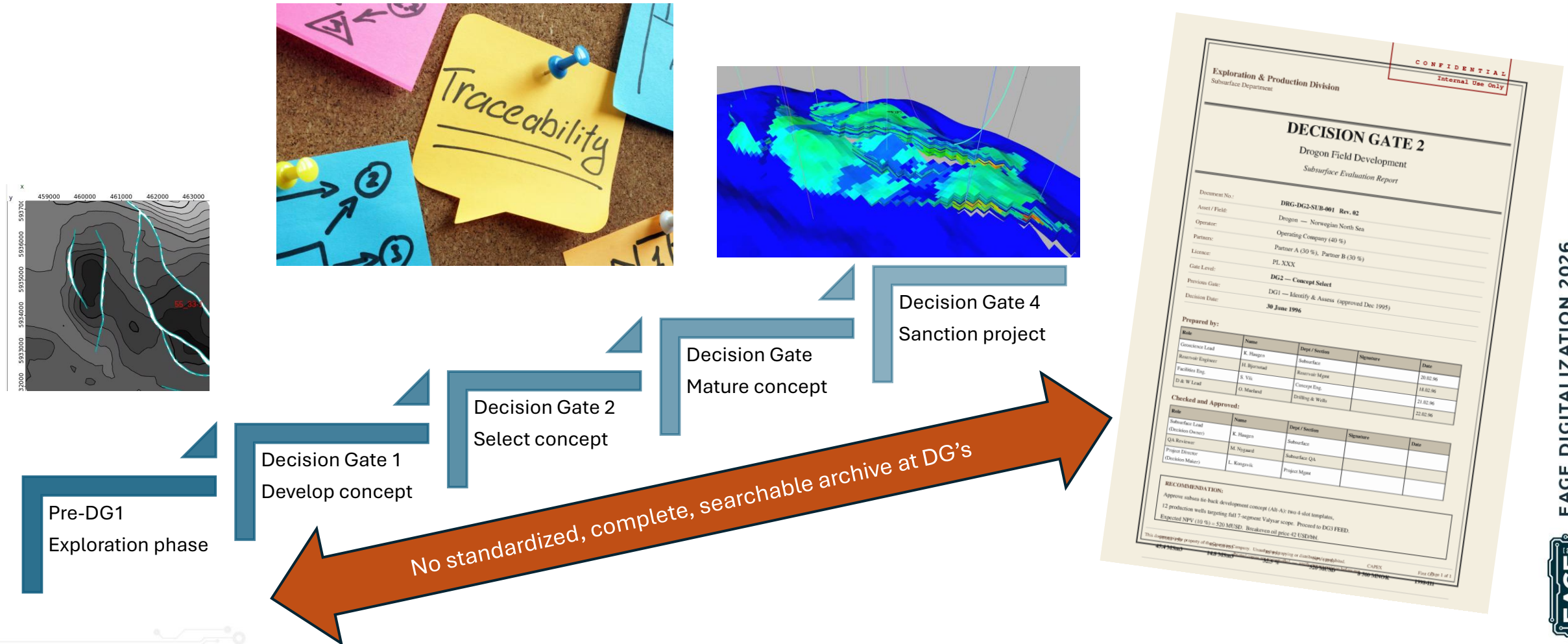
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Subsurface Decision Gates

The past: fragmented Data Management



Impact

Current Challenges in Decision Gate Data Management

Challenge	Current state	Impact
Data Storage Revise the historical decision with added data!	Single system pre-DG1 but fragmented beyond	Loss of continuity across DGs: cannot reproduce decisions if activities not recorded
Traceability What drove a volume drop between gates?	No formal linkage of inputs-outputs.	No coherency in attribute/volumetric changes, manual comparison error-prone, not auditable
Searchability Query all evidence used for DG2!	No standardized archive and poor relationships	Limited ability to query and analyze past decisions
Uncertainty and Performance Tracking How can we avoid poor production attainment?	Partial ensemble-based lifecycle record, no consolidated risk evolution record	Limited uncertainty evaluation reduction over time; prevents performance analysis

The Strategy

OSDU as Decision-Gate Backbone

Performance Tracking

Compare volumetric estimates across DG1–DGx and attribute changes to input parameters.

Uncertainty Calibration

Discover data relationships; refine input distributions for improved decisions

Risk Governance

Link risks and mitigations to decisions for audit readiness.

Evidence Consolidation

Package subsurface modelling input & output, concepts, and documents under BusinessDecision

“Drogon” demo objective

Collaborate

System of Record

- Prove OSDU can serve as the canonical backbone for decision-gate data
- Demonstrate relevant data types, schemas, and reference data and entity relationship support.
- Reference to complete persisted decision data: OSDU Reservoir DDMS DataSpace with geomodel and interpretation data and parameters, Catalog with tables, documents, etc.
- Enable rich query; use relationships, and granularity across gates
- Generate Report from BusinessDecision metadata and data such as volumes, profiles, maps

Risk Governance

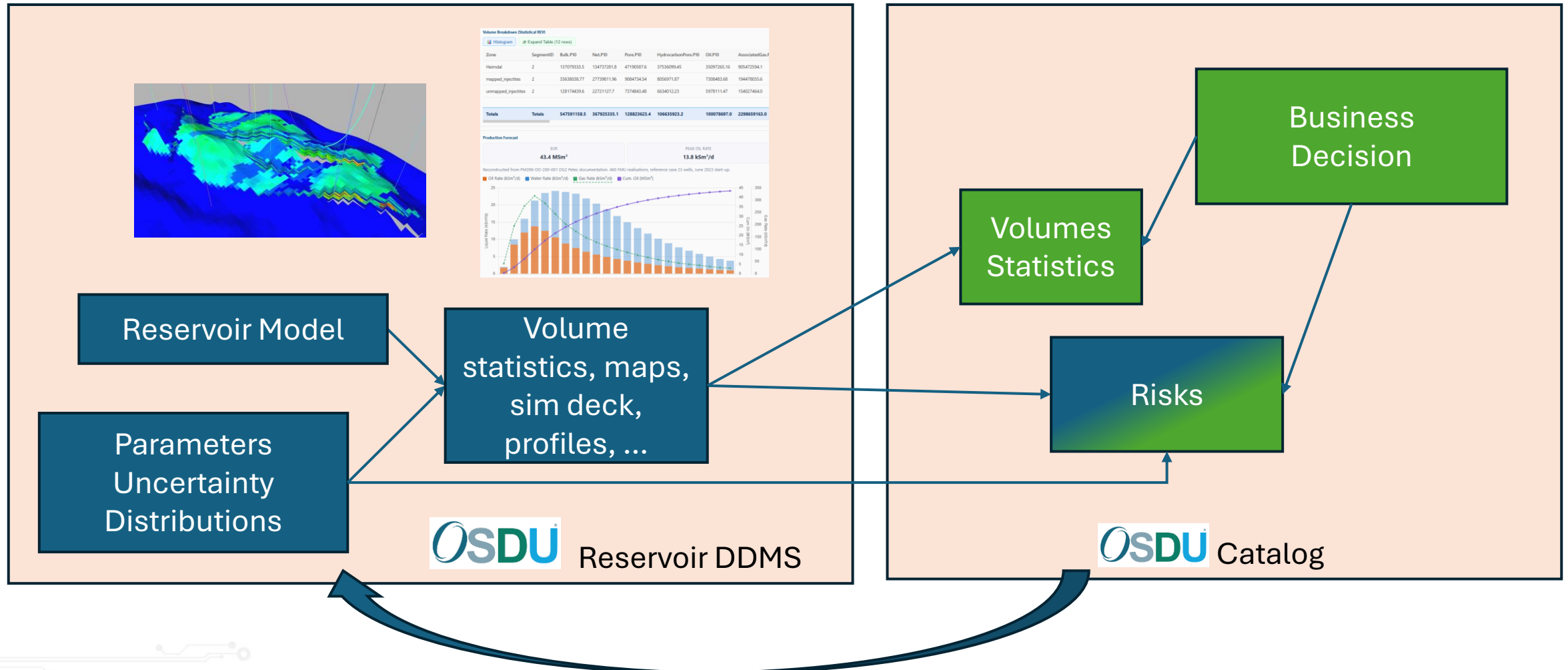
- Link risks and mitigations to decisions and uncertainty parameters, audit ready.
- Track risk and uncertainty evolution from DG0 onwards.

Uncertainty Calibration

- Use outputs to discover data relationships, refine input distributions for improved future field development decisions.
- Monitor changes across decision gates
- Storage of uncertainty distributions/parameters Activity with Business Decisions.

Demo workflow

Reservoir Data in Business Decisions ... simplified



Building blocks

Commonly used kinds of records

Master-data, unique valid keys for all DG

Reservoir: reservoir entity of interest.

ReservoirSegment — segments under the reservoir (compartments like fault block, zone, fluid region)

BusinessDecision – one per gate

Risk

Reference-data, examples for canonical types

UnitsOfMeasure

Statistics facets: P10,P50,P90, Mean, ...

Canonical volume property types: Bulk,Net,Pore,Oil,AssociatedGas, ...

Work Product Components (WPC), derived objects per DG – linked to Master Data

Design Matrix, Time series etc.: ColumnBasedTable.

Spatial data: grids, properties, as WPCs (e.g., STOIIP 2dGrid map), GeoLabelSet

Volumes: ReservoirEstimatedVolumes, per-realization and statistics

Activity: Use Parameters[] with ParameterRole {input, output, context} to record run inputs/outputs and workflow context

Collections for DG input/output/scenarios/cases

PersistedCollection: set of referenced WPC

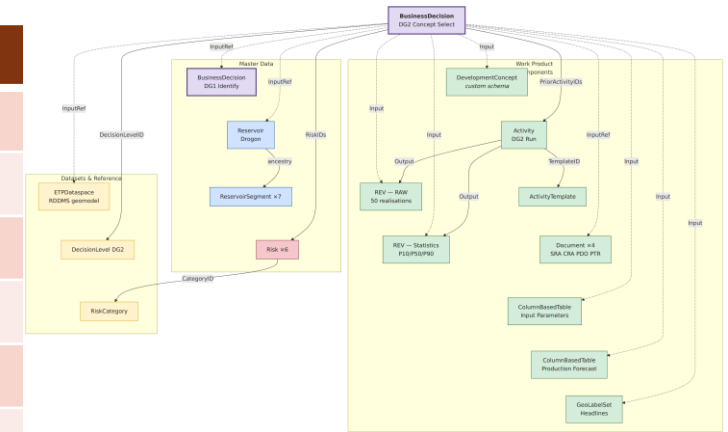
WorkProduct: covering versioned package: yearly revisions, PUD, DG ... (design/input, Activities, outputs).

CollaborationProjectCollection: curated working set while iterating – System of Engagement view

What OSDU provides

Drogon DG2 canonical schemas (~25 linked records)

Category	OSDU Kind	Purpose
Master	BusinessDecision	DG2 decision — central hub
Master	Reservoir	Drogon (shared with DG1)
Master	ReservoirSegment	7 segments
Master	Risk	6 DG2 risks
WPC	ReservoirEstimatedVolumes	Realisations + P10/P50/P90
WPC	ColumnBasedTable	Input params + production forecast
WPC	Activity + ActivityTemplate	Workflow run + template, captures methods input&output
WPC	Document	SRA, CRA, PDO, ...
WPC	GeoLabelSet	Headline volumes per segment
Custom	DevelopmentConcept	Custom LOCAL schema
Dataset	ETPDataspace	Reservoir DDMS DataSpace pointer: here spatial reservoir model is stored
Ref	DecisionLevel / RiskCategory / ... UoM, FacetRoles, PropertyTypes	Reference catalogs



- Masterdata (Reservoir, Segments) are shared across gates. WPCs (volumes, risks, concepts) exist per gate. BusinessDecision links them.
- Relationships are handled via Parameters[], RiskIDs, PriorActivityIDs, Ancestry. E.g., OSDU separates searchable metadata in storage from complex data in optimized Reservoir DDMS.

Drogon DG2 BusinessDecision WPC 1

- Search and analyse BusinessDecision related data in OSDU data platform
- Public github PoC demo incl. pipeline, data and client

Search <https://github.com/equinor/eageosdudrogon>

Kind (primary)

query

limit

returnedFields (comma separated)

Results (3)

Name	Description
GRAND — Decision Gate 2 DG2 Concept Select	DG2 concept selection for the Grane Northern Area Development (GRAND) project. The GRAND area comprises F-Structure and D-Structure c
Drogon — Decision Gate 2 DG2 Concept Select	DG2 Concept Select for the Drogon field development. Valysar fluvial formation across 7 reservoir segments (NorthSea, NorthHorst, CentralH
Drogon — Decision Gate 1 DG1 Identify & Assess	DG1 evaluation of the Valysar fluvial formation in the Drogon field, covering static in-place volume estimation across 7 reservoir segments (N

Select a result

Record:

Drogon — Decision Gate 2 DG2 Concept Select

Overview

Name Drogon — Decision Gate 2 DG2 Concept Select

Description DG2 Concept Select for the Drogon field development. Valysar fluvial formation across 7 reservoir segments (NorthSea, NorthHorst, CentralHorst, CentralFlanks, CentralSouth, SouthWing, EastLobe) with three facies types (Channel, Crevasse, Floodplain). Revised petrophysical interpretation: porosity reduced by factor 0.8 based on core analysis and thin sections. 50 FMU realisations quantify STOIIIP uncertainty: P90 33.8 / P50 45.4 / P10 59.4 MSm³ oil in-place. Recoverable oil (P50) 14.8 MSm³ at 32.5% recovery factor. Recommended concept: subsea tie-back to FPSO with 2×4-slot templates, 12 production wells. NPV@10% 520 MUSD, CAPEX 8,500 MNOK, first oil 2028-H1.

Kind osdu:wks:master-data--BusinessDecision:1.0.0

ID dev:master-data--BusinessDecision:Drogon-DG2-ConceptSelect:1

Version 1772376163257005

Drogon DG2 BusinessDecision WPC 2

- All key metadata and headline numbers are accessible
- All relying on OSDU well known schema
- Except local schema *DevelopmentConcept*

PROJECT

Drogon Field Development

DECISION GATE

DG2

STATUS

Pending

DECISION DATE

—

DUE DATE

2026-06-30

FORMATION

Valysar

WELLS

12

TARGET START-UP

2028-H1

HEADLINE VOLUMES (STOIIP)

STOIIP TOTAL

P90

P50

P10

11.55

10.93

9.85

MSm³

RECOVERABLE OIL

P90

P50

P10

3.47

3.77

3.84

MSm³

RECOVERY FACTOR

P90

P50

P10

30.0%

34.5%

39.0%

CENTRAL HORST

P90

P50

P10

5.93

4.91

4.52

MSm³

CENTRAL NORTH

P90

P50

P10

1.77

1.32

1.22

MSm³

CENTRAL RAMP

P90

P50

P10

0.68

0.53

0.51

MSm³

CENTRAL SOUTH

P90

P50

P10

3.69

3.07

2.77

MSm³

EAST LOWLAND

P90

P50

P10

0.03

0.03

0.02

MSm³

NORTH HORST

P90

P50

P10

0.10

0.09

0.08

MSm³

WEST LOWLAND

P90

P50

P10

0.48

0.37

0.31

MSm³

DEVELOPMENT CONCEPT

Subsea development with 2×4-slot templates (8 slots + 2 contingent), tie-back to converted FPSO via dual 10" production flowlines and 6" gas-lift line. Distance to FPSO ~8 km. Water depth 108 m. Valysar formation at ~1700 m TVD MSL.

WELLS

12

TEMPLATE SLOTS

10 (2 centres)

HOST FACILITY

Drogon FPSO (converted)

DISTANCE TO HOST

8 km

WATER DEPTH

108 m

FORMATION

Valysar

FIELD AREA

Drogon

TARGET START-UP

2028-H1

CONTINGENT WELLS

2

FLOWLINE

2×10" production + 6" gas lift

SUBSEA BOOST

Yes

WATER TREATMENT

5000 m³/d

INJECTION STRATEGY

Water injection for pressure support (4 injectors planned Phase 2)

WELL PLAN

12 prod. 4 inj. (Ph2) ~3200 m MD 45 d/well Frac-pack + ICD lower completion

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ENHANCING PREDICTIONS & INVESTMENTS WITH DIGITAL TECHNOLOGIES

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DIGITAL

Drogon DG2 BusinessDecision WPC 3

- All key metadata and headline numbers are accessible
- All relying on OSDU well known schema
- Except local schema *DevelopmentConcept*

RESERVOIR PROPERTIES

PRESSURE

170 bar

TEMPERATURE

72 °C

NET / GROSS

0.85

POROSITY (CHANNEL)

0.28

POROSITY (CREVASSE)

0.21

POROSITY (FLOODPLAIN)

0.1

AVG. PERMEABILITY

450 mD

OWC (TVDSS)

1710 m

OIL VISCOSITY

1.2 cP

BO FACTOR

1.12

GOR

85 Sm³/Sm³

KEY ECONOMICS

NPV 10PCT

\$520 MUSD

IRR

17%

CAPEX

8500 MNOK

OPEX PA

420 MNOK

BREAKEVEN OIL

\$42 USDperbbl

PAYBACK

7.0 years

SCHEDULE

DG2 Concept Select

2026-02-28

Completed

DG3 FEED

2027-01-01

Planned

FID / DG4

2027-07-01

Planned

FPSO Drydock Start

2027-10-01

Planned

Subsea Installation

2028-01-01

Planned

First Oil

2028-06-01

Planned

Plateau Production

2029-01-01

Planned

ALTERNATIVES

Alt-A: Subsea tie-back to FPSO (2×4-slot templates)

Rank 1

Approve

Highest NPV; leverages planned FPSO conversion for Drogon area. 12 wells across 7 segments, 2×4-slot templates with 2 contingent slots for infill. Subsea boosting pump included for low-pressure phase. Modular water treatment allows staged capacity expansion.

NPV \$520M

CAPEX 8500 MNOK

IRR 17%

Alt-B: Reduced scope — CentralHorst + CentralSouth only

Rank 2

Consider

Focus on the two highest-confidence segments (Channel facies dominant, lowest porosity uncertainty). 6 wells, single 4-slot template + 2 satellite wells. Lower investment but captures only ~55% of recoverable resource. Provides first-oil option if FPSO schedule slips.

NPV \$300M

CAPEX 4800 MNOK

IRR 20%

Alt-C: Defer — acquire DG3-quality appraisal data

Rank 3

Fallback

NorthHorst and EastLobe fault compartmentalisation remains poorly constrained. Additional well test data (est. 12 months, 200 MNOK) would reduce dynamic uncertainty range by ~40%. Risk of losing FPSO drydock window if deferred beyond 2026-Q3.

Drogon DG2 BusinessDecision WPC 4

- Link Risk master data
- Link to the DataSpace in Reservoir DDMS where the Geomodel input and output is archived, incl. parameters/Activity to be reproducible

RISKS

Drogon DG2 — Porosity and cementation uncertainty

Drogon DG2 — Fault transmissibility and reservoir compartmentalization

Drogon DG2 — HSE and environmental impact

Drogon DG2 — Schedule risk (FPSO and long-lead equipment)

Drogon-DG2-OWCDepth

Drogon-DG2-RecoveryFactor

UNCERTAINTY SUMMARY ▸ ▾

FMU static + dynamic uncertainty from 50 geo-realizations across 7 segments and 3 facies types. Porosity revised $\times 0.8$ from DG1. Dynamic forecasts run on 3 selected realizations (P90/P50/P10) with 5 sensitivity cases each.

Method: FMU Level 3 static + dynamic uncertainty workflow (Valysar geomodel v2)

50 realisations

P90: Real 42

P50: Real 1

P10: Real 17

See stat WPC for full P10/P50/P90 breakdown per segment & facies. Recoverable estimated from dynamic simulation with 12-producer base case.

Metric	P90	P50	P10
STOIIP (MSm ³)	33.8	45.4	59.4
Recoverable (MSm ³)	10.0	14.8	20.6
Recovery Factor (%)	28.0	32.5	37.0

INPUT PARAMETERS ▸

DATA SOURCES

GeoModelDataspace Reservoir DDMS

RDDMS ETP dataspace with the Drogon DG2 geomodel EPC files

OSDU: dev:dataset--ETPDataspace:maap-drogon_dg:1

AUTHORS & GOVERNANCE ▸

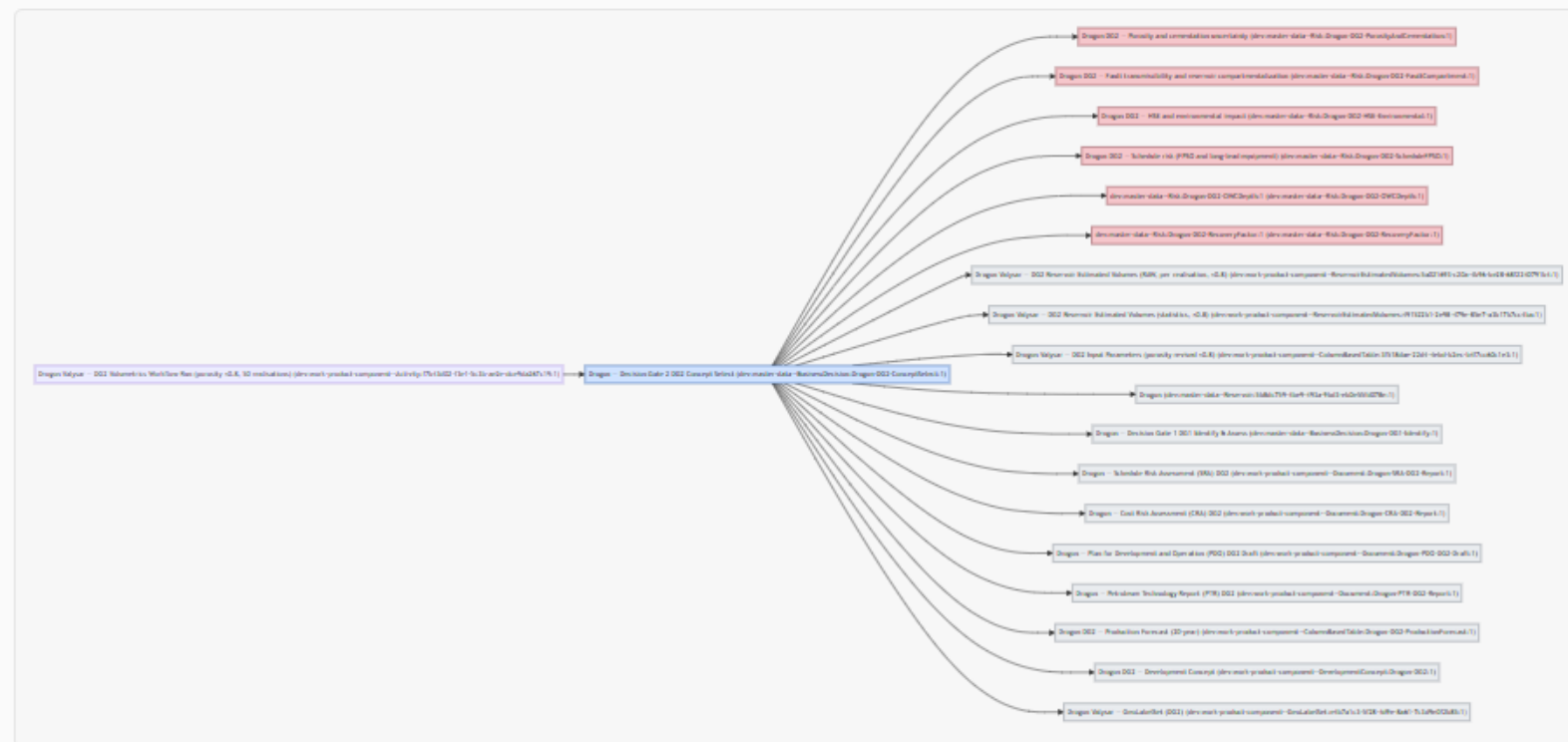
RECOMMENDATIONS FOR NEXT GATE

1. Execute FEED for FPSO conversion and subsea installation scope
2. Secure FPSO drydock slot (2027-Q4 target) with dual-yard tendering strategy
3. Drill NorthHorst appraisal sidetrack to constrain fault compartmentalisation model
4. Finalise Phase 2 water injection well locations based on DG3 dynamic simulation
5. Complete EIA with cold-water coral avoidance routing for flowline corridor
6. Run Level 3+ FMU workflow with 100+ realisations for DG3 volumetric basis
7. Prepare PDO final draft for MPE submission post-FID

Drogon DG2 BusinessDecision WPC 5

- BusinessDecision is a rich repository, with link to all required documents and data

Relationships



Linked objects

- Drogon DG2 — Porosity and cementation uncertainty (risk)
- Drogon DG2 — Fault transmissibility and reservoir compartmentalization (risk)
- Drogon DG2 — HSE and environmental impact (risk)
- Drogon DG2 — Schedule risk (FPSO and long-lead equipment) (risk)
- dev-master-data--Risk:Drogon-DG2-OWCDepth:1 (risk)
- dev-master-data--Risk:Drogon-DG2-RecoveryFactor:1 (risk)
- Drogon Valysar — DG2 Reservoir Estimated Volumes (RAW, per realisation, $\times 0.8$) (ref)
- Drogon Valysar — DG2 Reservoir Estimated Volumes (statistics, $\times 0.8$) (ref)
- Drogon Valysar — DG2 Input Parameters (porosity revised $\times 0.8$) (ref)
- Drogon (ref)
- Drogon — Decision Gate 1 DG1 Identify & Assess (ref)
- Drogon — Schedule Risk Assessment (SRA) DG2 (ref)
- Drogon — Cost Risk Assessment (CRA) DG2 (ref)
- Drogon — Plan for Development and Operation (PDO) DG2 Draft (ref)
- Drogon — Petroleum Technology Report (PTR) DG2 (ref)
- Drogon DG2 — Production Forecast (20-year) (ref)
- Drogon DG2 — Development Concept (ref)
- Drogon Valysar — GeoLabelSet (DG2) (ref)

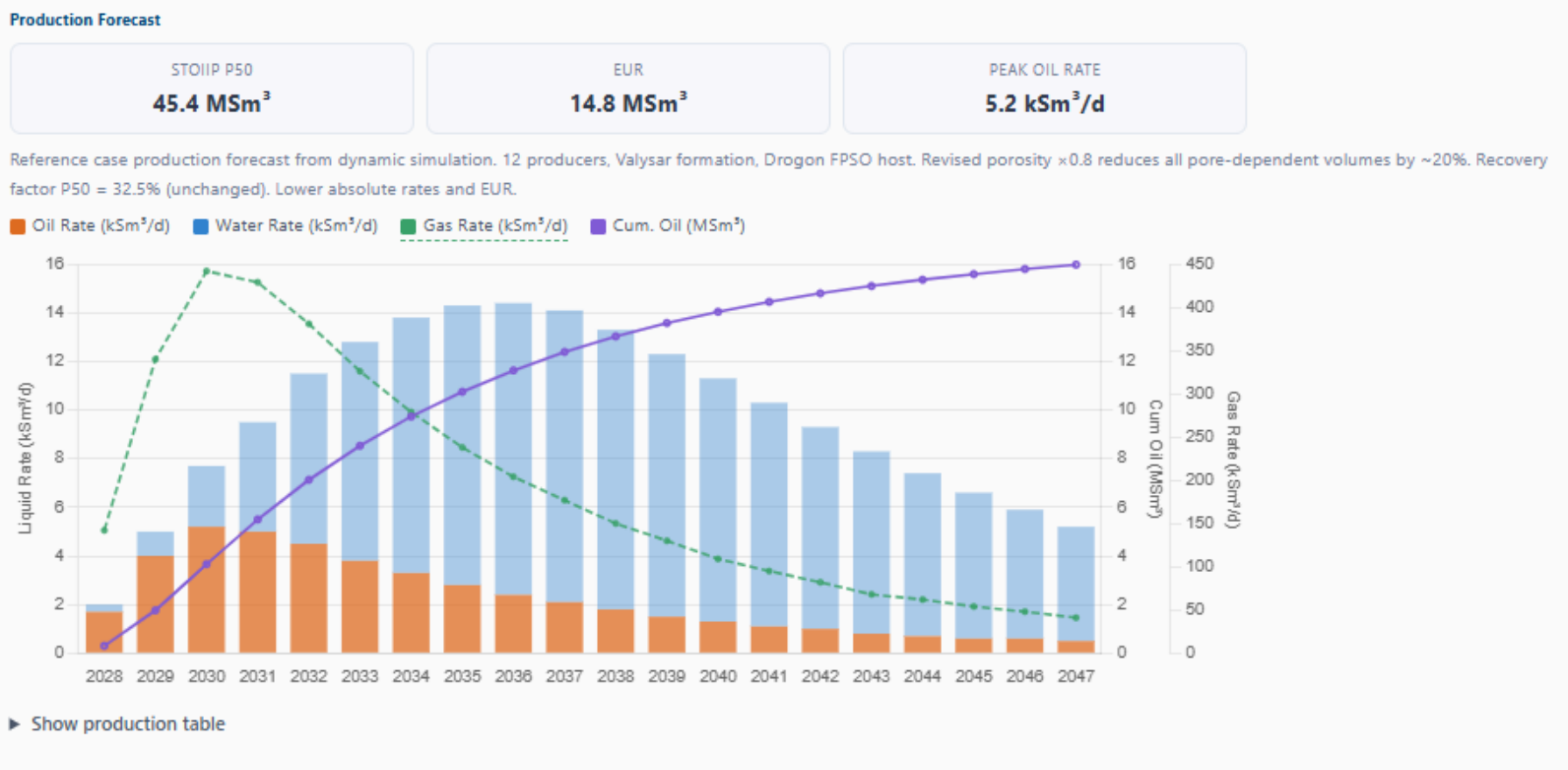
Drogon DG2 BusinessDecision WPC 6

- Via references, key data like volume statistics and profiles are available in a canonical way ...
- Let's use it for query and analysis!

Volume Breakdown (Statistical REV)

[Histogram](#) [Expand Table \(40 rows\)](#)

Zone	SegmentID	Facies	BulkOil.P10	PoreOil.P10	HydrocarbonPoreOil.P10	Oil.P10	AssociatedGas.P10	BulkGas.P10	Po
Valysar	Central Horst	Channel	24496296.6702	4881894.2412	4597227.4706	3303258.1546	425063794.0094	0.0	0.0
Valysar	Central Horst	Coal	10136266.6985	0.0	0.0	0.0	0.0	0.0	0.0
Valysar	Central Horst	Crevasse	7834596.3103	1140878.7826	842188.5574	614287.4895	83187681.4662	0.0	0.0
TOTAL	TOTAL	TOTAL	186997460.0491	21935009.838499997	14142406.5385	9852971.5993	1381556182.8033001	24146803.990799997	23



Analysis: DG comparison dashboard

Sensitivity Analysis (Tornado)

Vary one input at a time (OWC, porosity, NTG), hold others at P50. Rank by STOIP delta → tornado chart shows dominant drivers.

Design Matrix Correlation

Realisations × Input variables. Compute correlation between each input column and output STOIP → identify key levers.

Gate-to-Gate Delta Attribution

Compare DG1 vs DG2 Activity records. Diff input ranges and exp. design (properties, scenarios), attribute volume change.

What-If / Scenario Comparison

Use BD.Parameters[] to swap input WPCs. Re-run with alternative porosity model → compare recoverable volumes and economic impact.

Drogon Demo in OSDU:

Reservoir DDMS Dataspace

Activity → Parameter DesignMatrix

Activity → Variables, Methods

ColumnBasedTable → input parameters

REV RAW → Segment volumes

REV STAT → P10/P50/P90 ... per segment

Risk records → Severity/Probability
before & after

All queryable, all linked via
BusinessDecision.

→ Enough structure for automated
sensitivity & AI-generated insight
reports.

Analysis: DG comparison dashboard 1

Current Capabilities

- Reservoir selection
- Discovery of BDs referencing reservoir
- Gate-by-gate comparison (DG1 → DG2)
- Metric deltas: STOIIP, Recoverable, RF
- Economics: NPV, CAPEX, OPEX, IRR, breakeven
- GeoLabelSet enrichment per segment
- Risk tracking: severity/probability evolution
- Property delta computation (porosity, NTG ...)

Analyse — Reservoir Decision Gate Tracking

Select a reservoir to compare Business Decisions across decision gates. Track how input parameters, volumes, economics, and risks evolve from early screening to concept selection.

1 — Select Reservoir

*

Drogon — dev:master-data--Reservoir:5b8dc759-4be9-493a-9bd3-eb0e55fd078e:1
Grand — dev:master-data--Reservoir:f9585655-83d8-4549-ae3e-2dffc2cd5937:1

Loaded 2 decision gate(s).

Drogon

dev:master-data--Reservoir:5b8dc759-4be9-493a-9bd3-eb0e55fd078e:1

DG1

2026-02-28

Approved

DG2

2026-06-30

Pending

VOLUMES (MSM³)

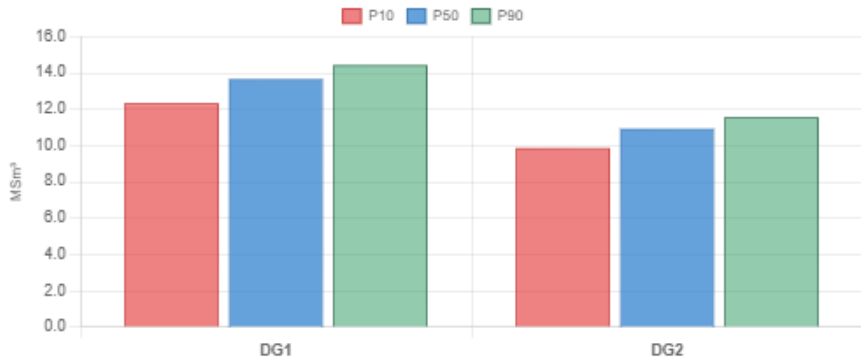
METRIC	DG1	DG2	Δ (LAST STEP)
STOIIP P90	14.4	11.6	-2.89 (-20%)
STOIIP P50	13.7	10.9	-2.73 (-20%)
STOIIP P10	12.3	9.85	-2.46 (-20%)
Recoverable P90	4.33	3.47	-0.866 (-20%)
Recoverable P50	4.71	3.77	-0.943 (-20%)
Recoverable P10	4.80	3.84	-0.961 (-20%)
Rec. Factor P90 (%)	30	30	0 (0%)
Rec. Factor P50 (%)	34.5	34.5	0 (0%)
Rec. Factor P10 (%)	39	39	0 (0%)

Analysis: DG comparison dashboard 2

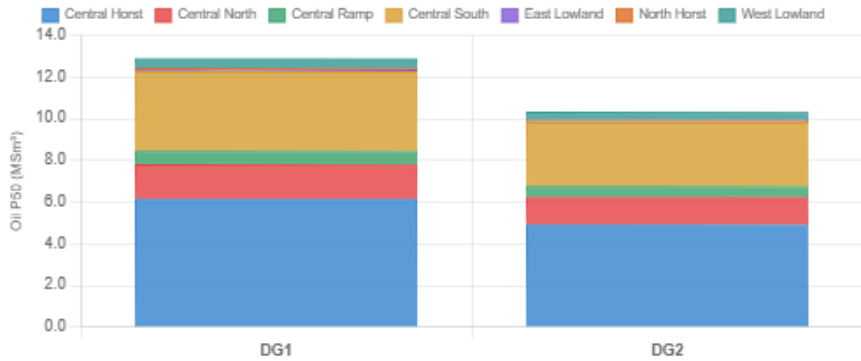
STOIIP BY SEGMENT — P50 (MSM³)

SEGMENT	DG1	DG2	Δ LAST
TOTAL	13.7	10.9	-2.73 (-20%)
Central Horst	6.14	4.91	-1.23 (-20%)
Central North	1.65	1.32	-0.330 (-20%)
Central Ramp	0.658	0.527	-0.132 (-20%)
Central South	3.84	3.07	-0.767 (-20%)
East Lowland	0.031	0.025	-0.006 (-20%)
North Horst	0.110	0.088	-0.022 (-20%)
West Lowland	0.466	0.373	-0.093 (-20%)

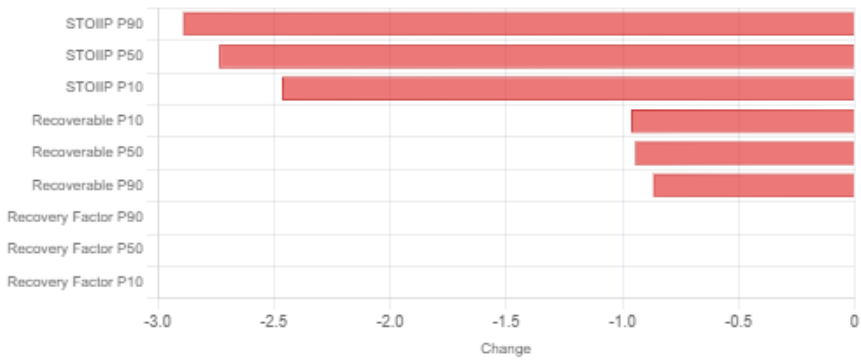
STOIIP UNCERTAINTY RANGE BY GATE



OIL IN PLACE BY SEGMENT



DELTA TORNADO — LAST GATE CHANGE



RISK EVOLUTION



Analysis: DG comparison dashboard 3

RESERVOIR PROPERTIES (GEOLABELSET)

PROPERTY	DG1	DG2	Δ LAST
FormationVolumeFactor	1.12	1.12	0 (0%)
GasOilRatio	85	85	0 (0%)
NetToGross	0.8500	0.8500	0 (0%)
OilWaterContact	1,710	1,710	0 (0%)
Permeability	450	450	0 (0%)
Porosity (Channel)	0.2800	0.2800	0 (0%)
Porosity (Crevasse)	0.2100	0.2100	0 (0%)
Porosity (Floodplain)	0.1000	0.1000	0 (0%)
Pressure	170	170	0 (0%)
Temperature	72	72	0 (0%)
Viscosity	1.20	1.20	0 (0%)

RISK EVOLUTION

GATE	RISKS	CHANGES
DG1	Drogon — Porosity and cementation uncertainty S2/P3 Drogon — Fault transmissibility and reservoir compartmentalization S3/P2 dev:master-data--RiskDrogon-OWCDepth:1	Baseline
DG2	Drogon DG2 — Porosity and cementation uncertainty S2/P2 ✓ Drogon DG2 — Fault transmissibility and reservoir compartmentalization S2/P2 ✓ Drogon DG2 — HSE and environmental impact S3/P1 ✓ Drogon DG2 — Schedule risk (FPSO and long-lead equipment) S2/P2 dev:master-data--RiskDrogon-DG2-OWCDepth:1 dev:master-data--RiskDrogon-DG2-RecoveryFactor:1	Fault transmissibility and reservoir compartmentalization: ✓ Mitigated Porosity and cementation uncertainty: ✓ Mitigated + dev:master-data--RiskDrogon-DG2-OWCDepth:1 + dev:master-data--RiskDrogon-DG2-RecoveryFactor:1 + Drogon DG2 — HSE and environmental impact + Drogon DG2 — Schedule risk (FPSO and long-lead equipment) —dev:master-data--RiskDrogon-OWCDepth:1

Synthesis — Cross-Gate Change Analysis

- STOIIP P50 decreased by 20% (-2.73 MSm³) from DG1 to DG2.
- Recoverable oil P50 decreased by 20% from DG1 to DG2.
- 2 risk(s) mitigated, 1 risk(s) removed, 4 new risk(s) identified between DG1 and DG2.
- Mitigation focus for next gate: address new risks and validate subsurface model assumptions to reduce volume uncertainty.

OSDU enabled traceability

Reservoir DDMS supports Business Decisions

OSDU RDDMS transforms fragmented, non-auditable workflows into a **transparent, traceable decision pipeline** supporting continuous performance improvement.

- Compare volumetric estimates across DG
- Attribute changes to underlying inputs
- Track uncertainty reduction and decision quality
- Enable repeatable, data-driven governance
- Maintain a long-term, searchable archive

Enhanced performance and maturation tracking

- Lifecycle Feedback Loop

Supports lifecycle analysis

- Searchable Archive

Full traceability and auditability

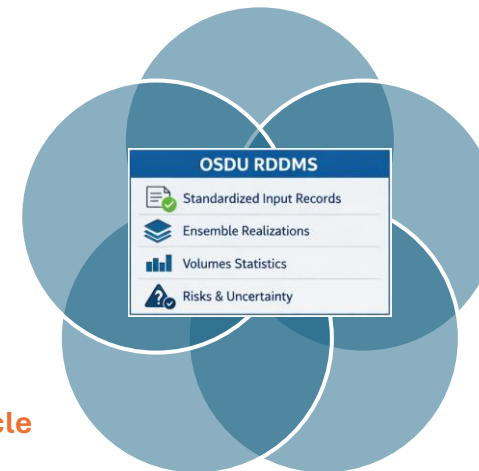
- Lossless Input/Output Records

Improved uncertainty tracking

- Ensemble-Based Data Support

Enable comparisons and analytics

- Standardized Data Models





We want to thank Equinor and Emerson-AspenTech for allowing us to present this work.

